PXL – Digital
421280 Software Analysis

System & System Context – Domain Model

Week 05 – period 01

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Content

- Problems & context of requirements gathering
- Some definitions
  - Object, class structure, attributes, operations
- Definition of a domain model
- Example of a domain & a domain model
- Characteristics and benefits of a domain model
- Definition of a class diagram
- The process of domain modeling
- General steps in domain modeling
- Developing and documenting Domain Models

Problems & context of requirements gathering

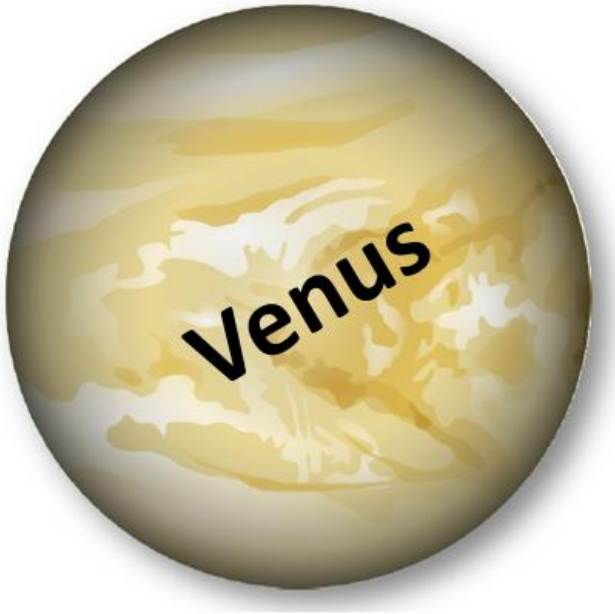
- Unknown knowledge field
- Unknown jargon
- Unknown problem
- Unknown ….

➔ Unknown problem domain

➔ Ambiguous communication

➔ A lot of communication

Problems & context of requirements gathering



Client

Developers

Other Developers

Definition of a domain model

"A domain model is like a map of the real world that your app will live in.

It shows which things exist, how they connect, and what rules apply — so your code later makes sense."

Definition of a domain model

- Captures the most important types of objects in a system
- Describing "things" in a system and how these things are related to each other
- A "thing" can be an object, a class, an interface, a package or a subsystem, which is part of the system being developed
- Very important process because it is employed throughout the entire system development life cycle
- **Thus ➔ Model of the problem domain with**
  - **Glossary of terms**
  - **Domain Objects – domain classes**
  - **Relations between terms**

Domain model: a little story …

- Students and staff at PXL can borrow bikes to move between campuses.
- Each bike has a unique ID and is parked at a station.
- Users can reserve a bike (max 3) through the app and return it to any station. There are 10 stations available.
- The system must record reservations, bike status, and usage history.

Domain model: a little story …

- Exercise
  - Find the key concepts (nouns)
    - Read text and highlight important things (nouns)
  - Define relationships (verbs)
    - How concepts connect: “Who does what with what?”
  - Add attributes (details):
    - Brainstorm what info each concept needs
  - Check business rules
    - Kind of relation? Multiplicities?
  - Draw and discuss
    - For now boxes and lines

Domain model: a little story …

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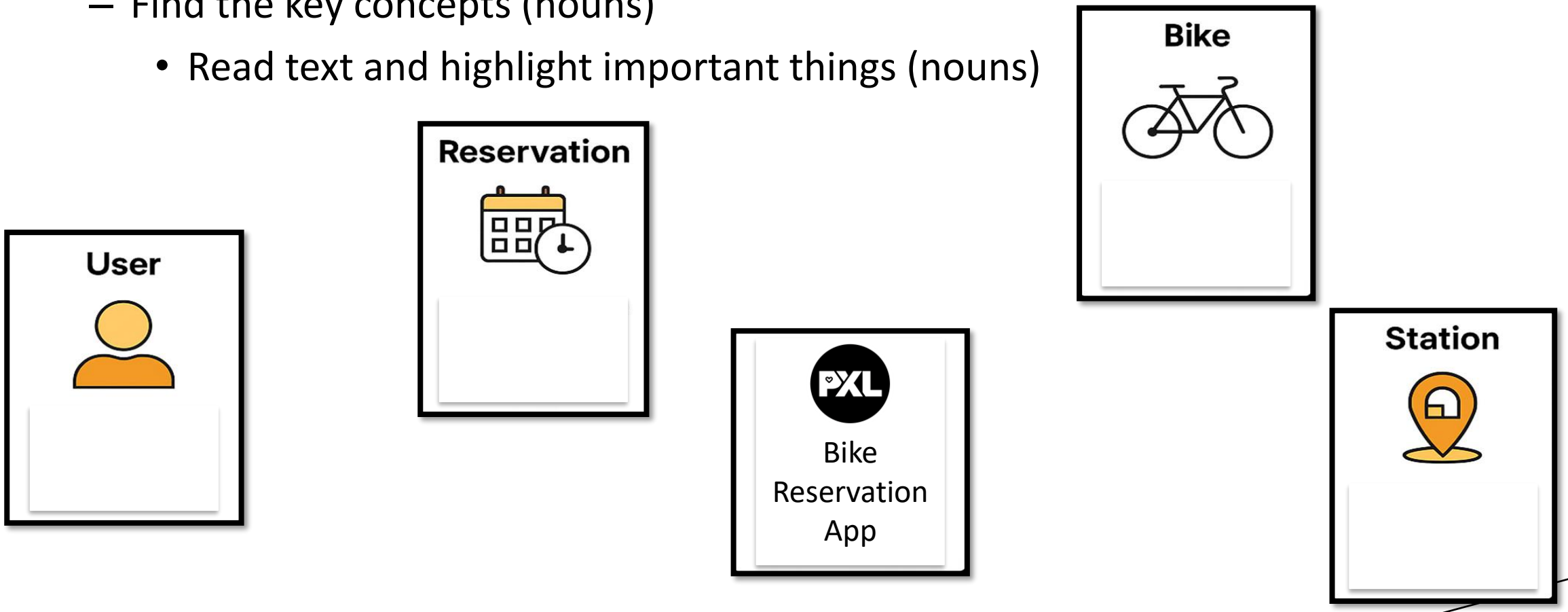

Domain model: a little story …

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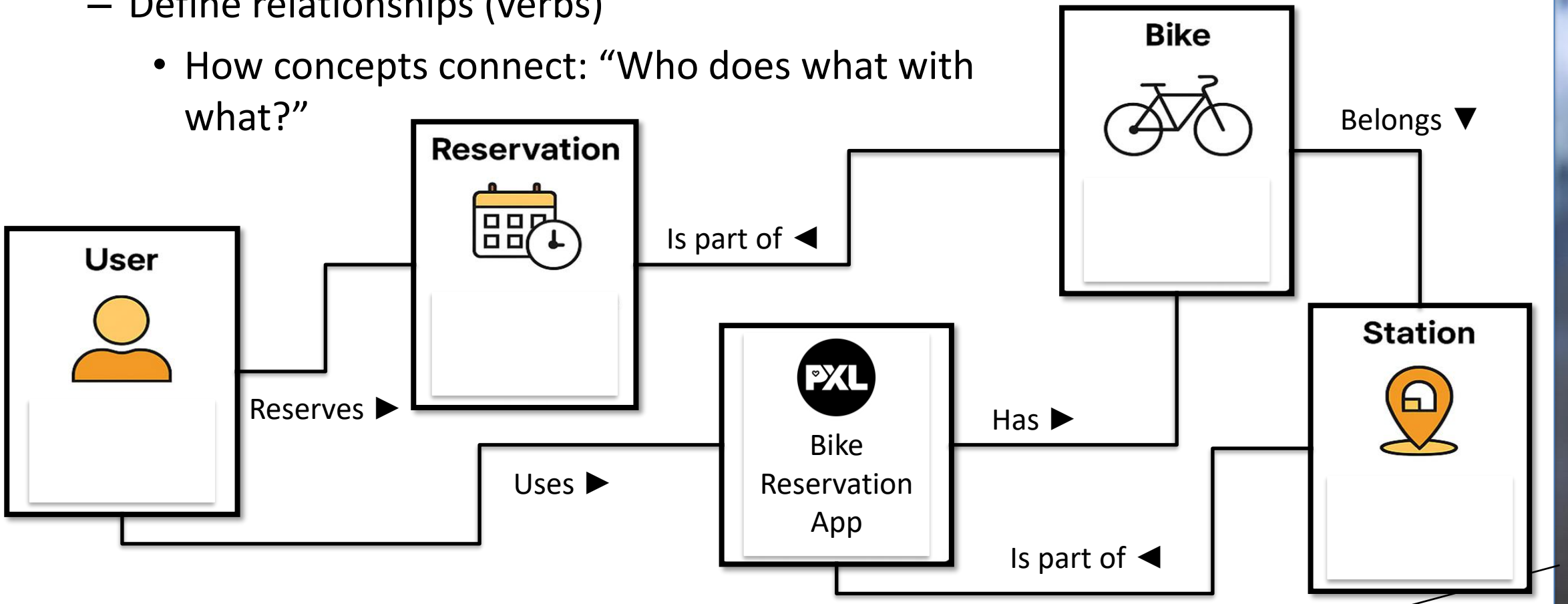

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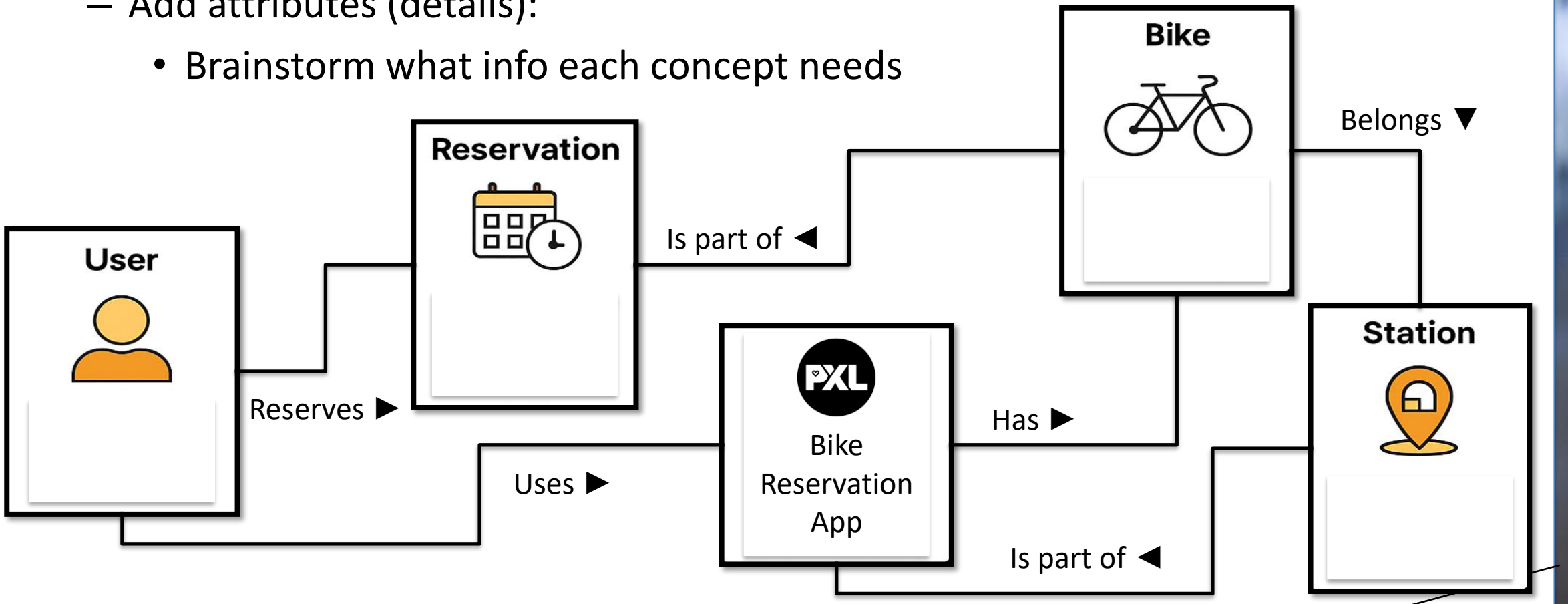

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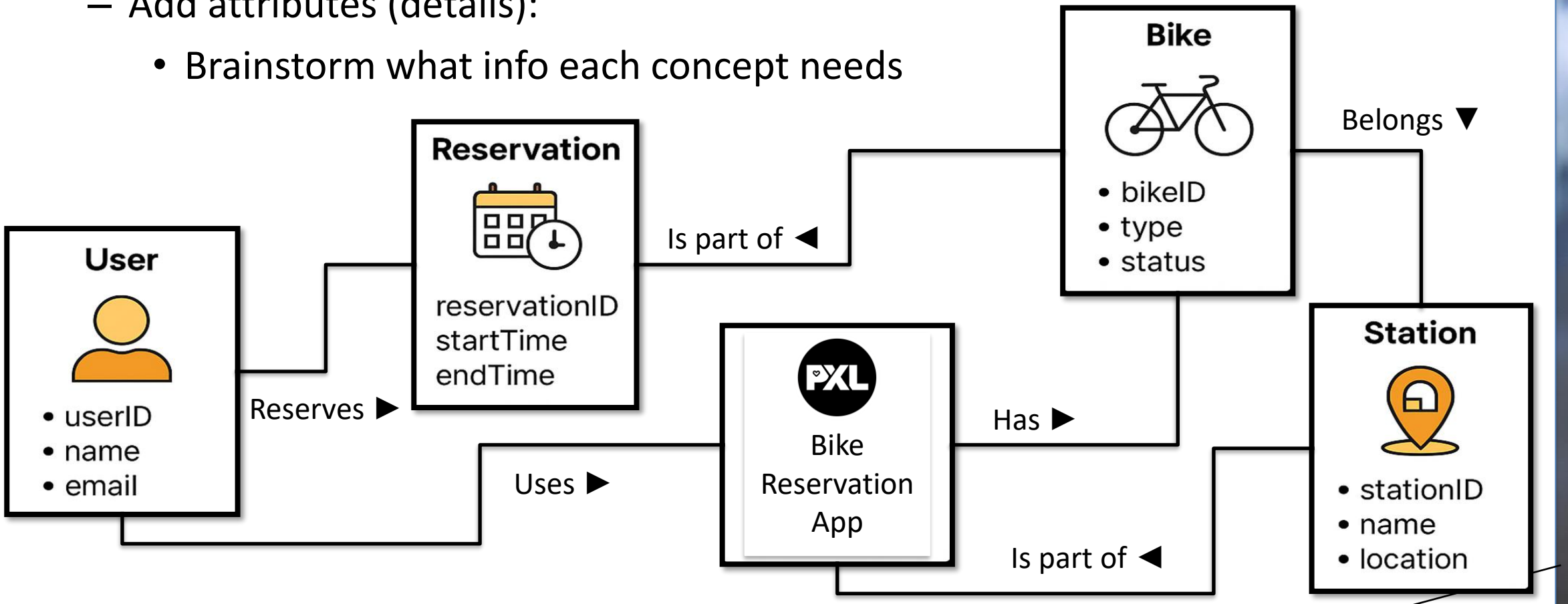

Domain model: a little story …

- Exercise
  - Check business rules
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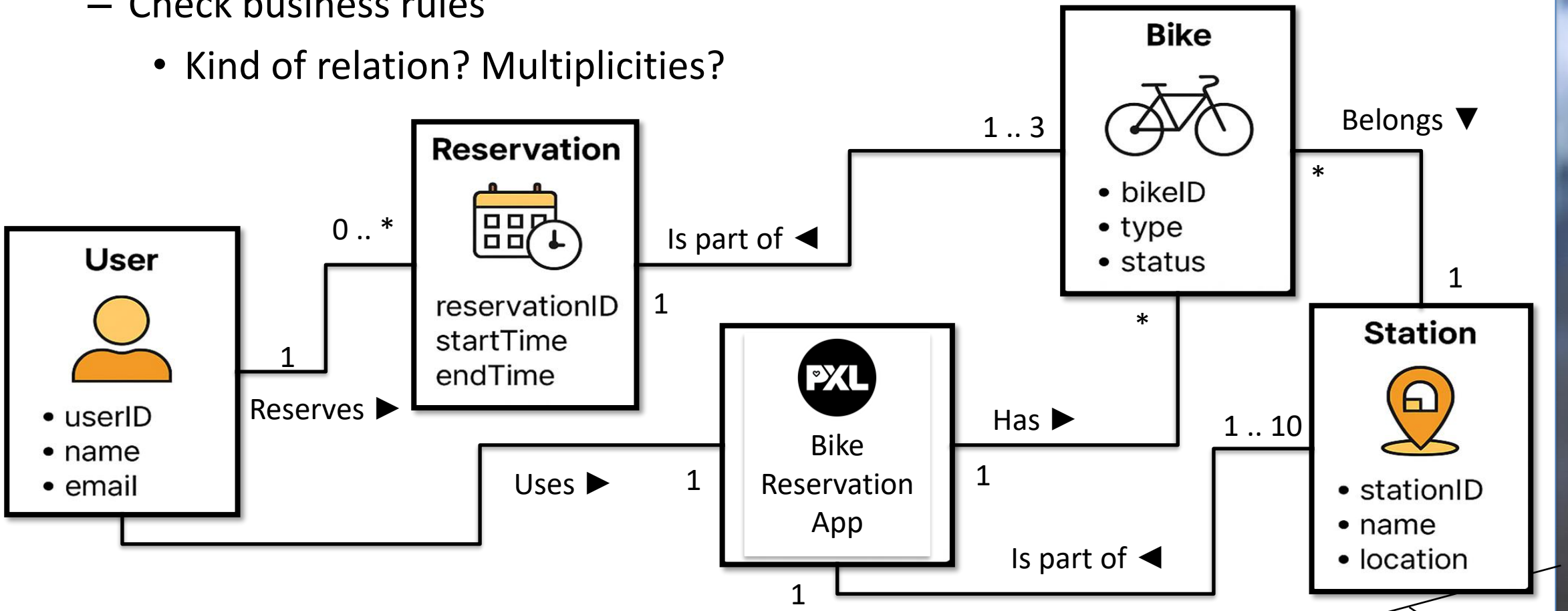
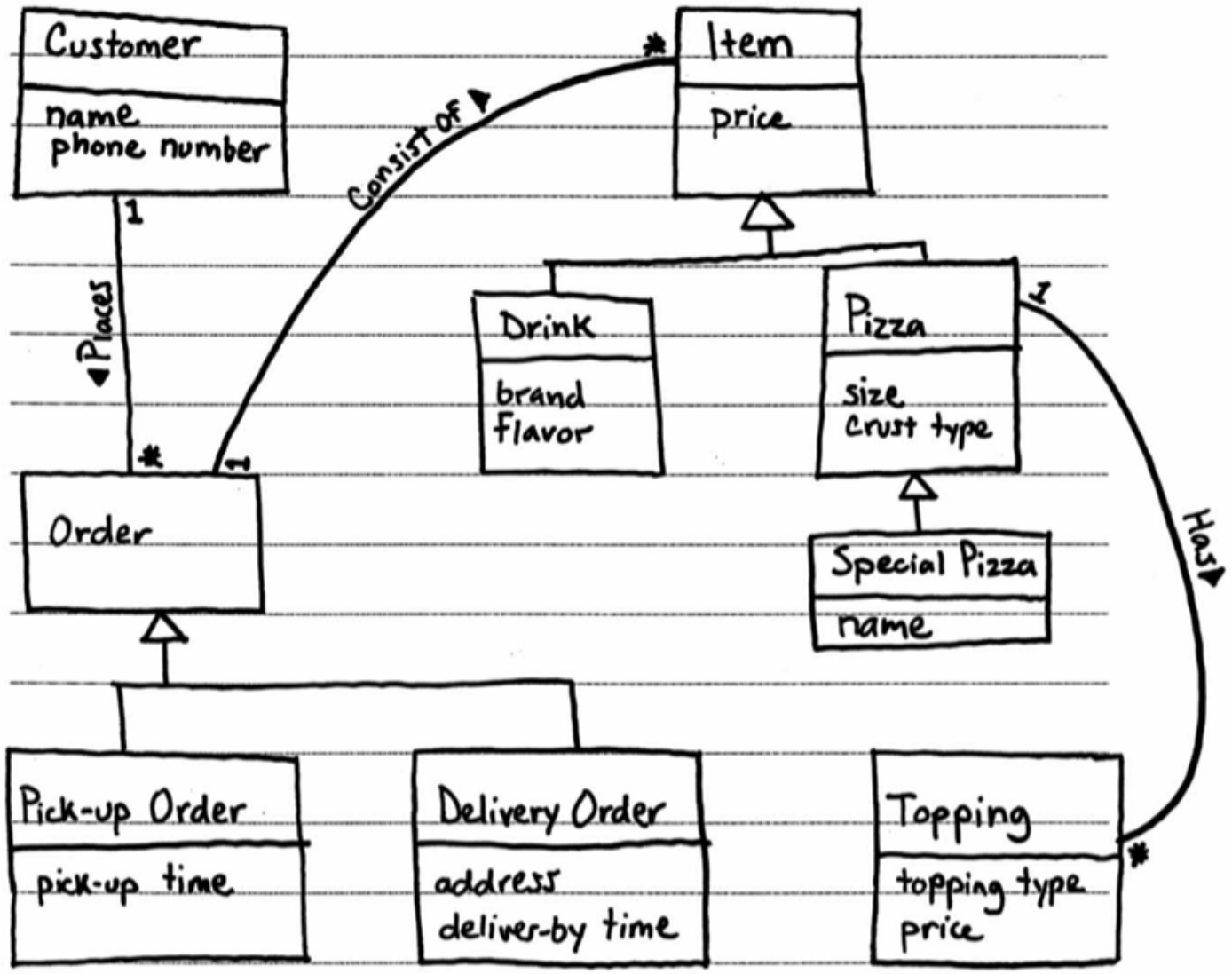


Another example of a domain model ➔ conceptual classes



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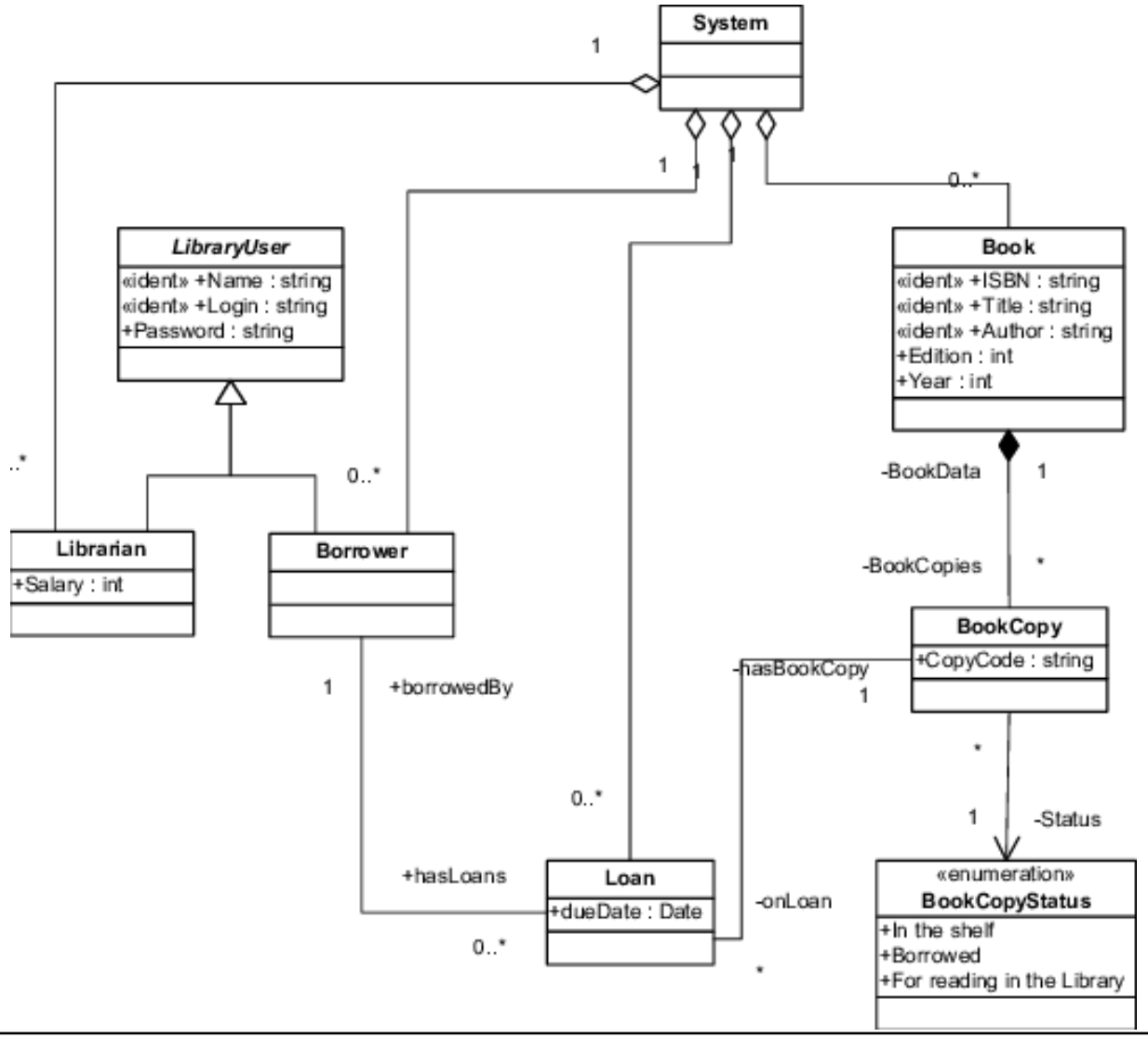

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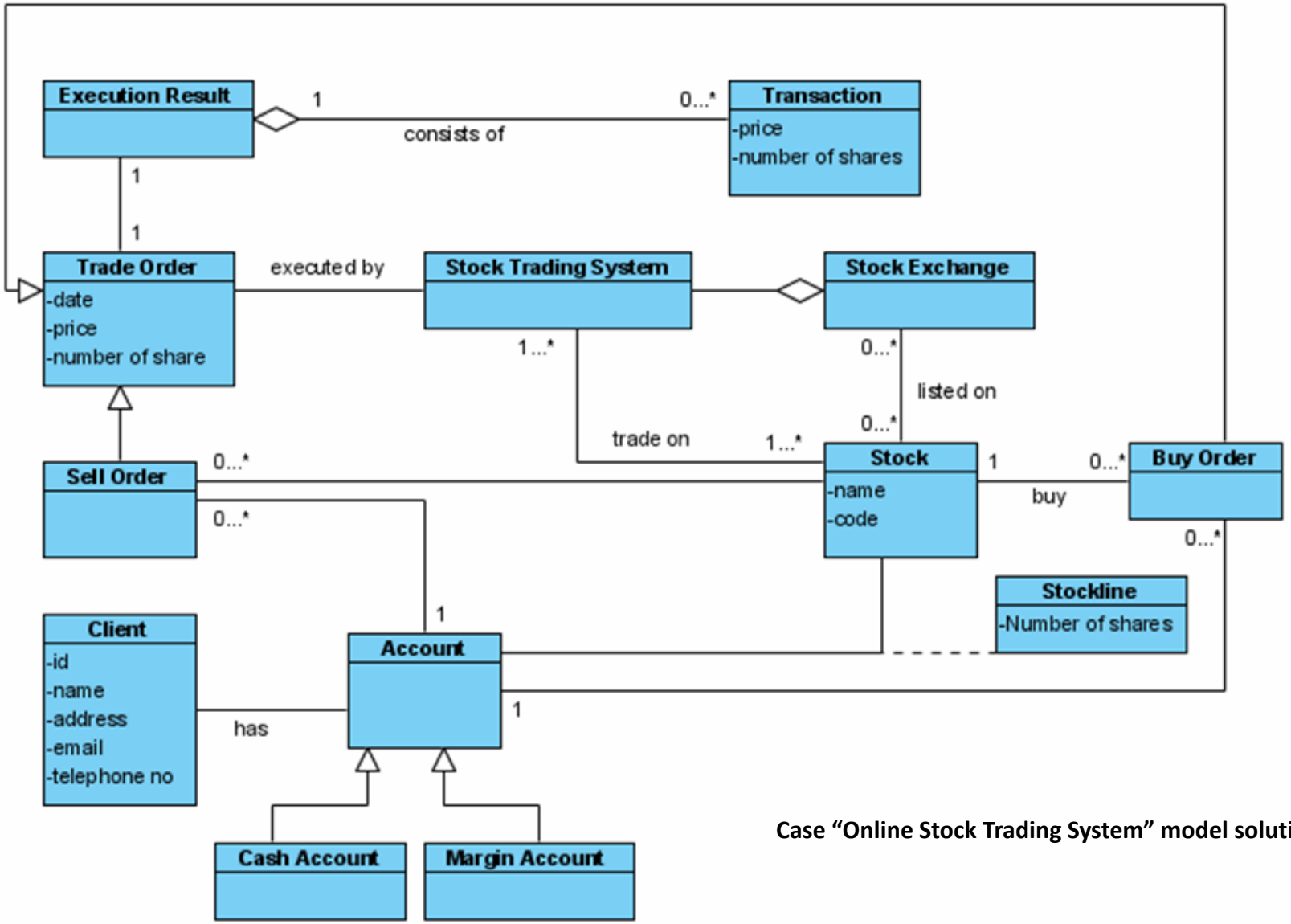


Case "Online Stock Trading System" model solution

Characteristics & benefits of a domain model (1)

- System of **abstractions** describing selected aspects of sphere of knowledge, influence, or activity
- Is **visual representation** of **meaningful real-world concepts**, **conceptual classes** pertinent to the domain needing to be modeled in software
- Thus: identify & relate **key concepts** in a domain
  - Also called "conceptual modeling"
- Shows **associations and relationships** between concepts
  - E.g., Payment PAYS-FOR Sales
- Shows **attributes** for information content
  - E.g. Sale records DATE and TIME

Characteristics & benefits of a domain model (2)

- Concepts of domain model: **data** involved in the business and business rules used in relation to that data
- Used **to solve problems** related to that domain, business
- Generally, uses **vocabulary of the domain** so that representation of the model can be used to **communicate with non-technical stakeholders**
- **Does not include operations / functions**
- **Does not describe software classes**
- **Does not describe software responsibilities**
- **Part of Object-Oriented Analysis**
  - **i.e. analysis of the problem space**

Definition of a class diagram

- A diagram that describe structure of system by showing the system's:
  - classes
  - its attributes and operations
  - relationships between classes

The process of domain modeling – Why?

- **Why domain modeling?**
  - You may **not know** the **domain** well
    - Details matter!
    - E.g.: does every student have exactly one major?
  - You don't want to forget **key concepts**
    - E.g.: a student's home college affects registration
  - You want to **agree** on **common set of terms**
    - E.g.: a freshman/sophomore vs. first-year/second-year
  - **Prepare to design**
    - Domain concepts are good candidates for OO classes

The process of domain modeling – How (1)?

- A **recommended flow** for domain modeling is shown below in sequential order:
  - **1. Understand the Problem**

    Write a short problem statement in plain language.

    Ask: What problem does the system solve? Who is involved?

    ‼ **Example**:

    "Students can borrow bikes through an app to move between PXL campuses."
  - **2. Identify Key Concepts (Nouns)**

    Scan the story for things (nouns) that matter.

    Mark potential classes or objects

    ‼ **Example**:

    User, Bike, Reservation, Station.

💡 **Tip:**
- **Count the occurrences**
- **Put them in an alphabetic list**
- **Good candidates occur often**
- **Make a glossary of terms**
- **Create a first domain class diagram**

The process of domain modeling – How (2)?

- A **recommended flow** for domain modeling is shown below in sequential order:
  - **3. Build a Shared Vocabulary**

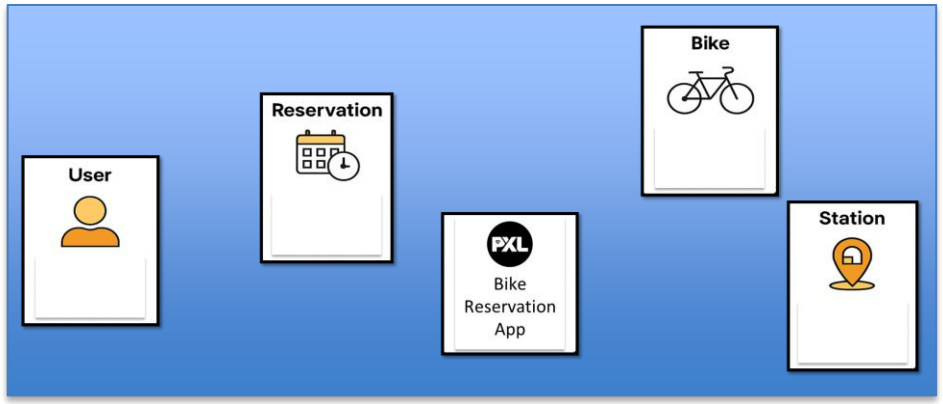
    Create a domain dictionary or glossary so everyone uses the same words.

    !! **Example**: Reservation = record of a user borrowing a bike

    Station = parking spot where bikes are stored
  - **4. Select and Define Core Domain Classes**

    Choose the most important concepts — the ones your system must manage.

    !! **Example**:



The process of domain modeling – How (3)?

- A **recommended flow** for domain modeling is shown below in sequential order:
  - **5. Sketch the First Domain Model**

    Draw boxes (classes) and lines (relationships).

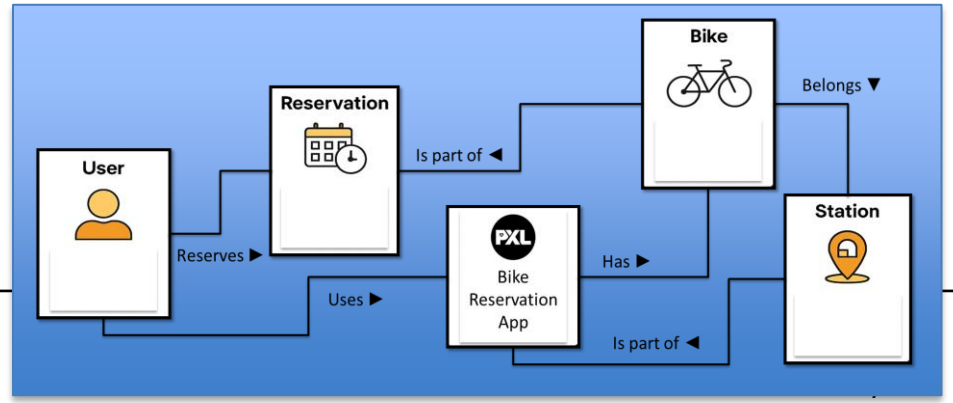
    Keep it rough — focus on meaning, not precision yet.
  - **6. Identify Associations**

    For each pair of classes, ask: “Do they interact or depend on each other?”

    Add arrows and relationship names (e.g. User reserves Bike).

    Bottom line - identify associations between concepts, including reading direction

    !! **Example**:



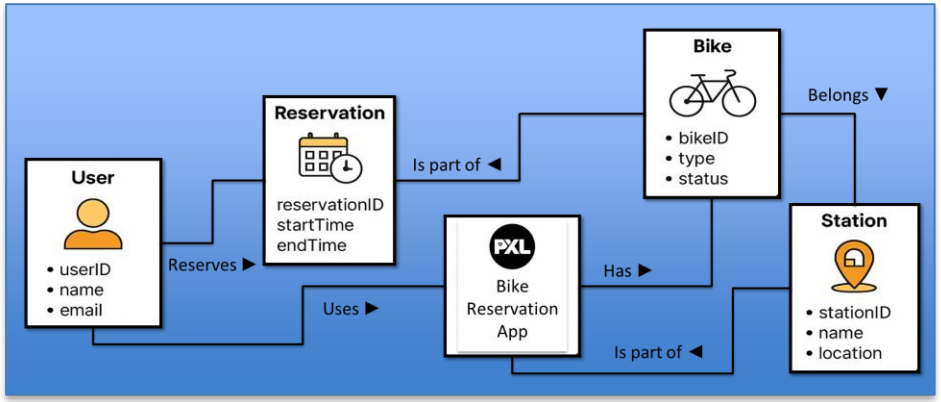
The process of domain modeling – How (4)?

- A **recommended flow** for domain modeling is shown below in sequential order:
  - **7. Add Attributes**

    Inside each class, list the essential data you need to know.

    ‼ **Example**:

    Bike: bikeID, status, type



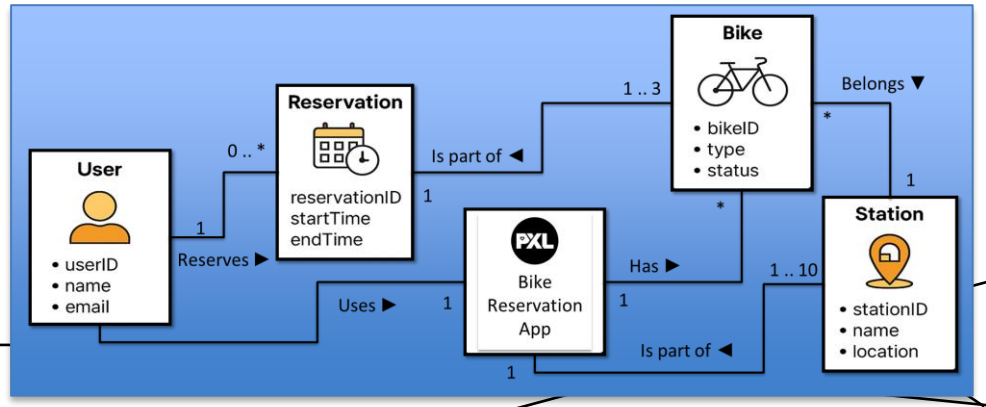
  - **8. Define Multiplicities**

    Specify how many objects are linked.

    ‼ **Example**:

    One User → many Reservations

    One Bike → one Station



Step 01 - Prepare problem statement

- Use interview notes of requirements
- Check for business use case descriptions
- Reuse existing models
- Gather other input information, first own notes of requirements
- Analyze typical use scenarios, analyze behavior
- Execute a brainstorming
- Collect first; organize, filter, and revise later

Step 02 – Identify concepts, classes & objects (1)

- Read descriptions, input documentation carefully
- Look for **nouns** ➔ **and underline these**
  - Words with definite and indefinite articles (“the”, “it“, “a”)
- Look for **verbs** because nouns execute these
  - Verbs indicate an action
- Look for **adjectives** because these tell something about the nouns
  - Characteristics, properties (=attributes)
  - Attributes (usually) derived from sentence structures as:
  - “X” has a “Y” and a “Z”
  - “X” is made up of a “Y” and a “Z”
  - “X” consists of a “Y” and a “Z”

Step 02 – Identify concepts, classes & objects (2)

- Other hint for identifying ➔use a category list
  - **Tangible things**: cars, telemetry data, terminals, classroom, playground, …
  - **Conceptual**: course, module, …
  - **Events**: landing, purchase, request, test, examination, seminar, …
  - **External organizations**: publisher, supplier, …
  - **Roles played**: mother, teacher, researcher, student, …
  - **Other system(s)**: admission system, grade reporting system, …
  - **Interactions**: loan, meeting, intersection, …
  - **Attributes**: cash balance, color, …
  - **Structure**, **devices**, **organizational units**, …

Step 02 – Identify concepts, classes & objects (3)

- Underlined words = domain concepts
- User does not count? Why?
  - Is an actor!
  - Underline but use another color

- Conclusion
  - “Noun” method is no guarantee for finding of all domain classes
  - In a lot of situations this is an ideal starting point for further investigation

Step 01 - Prepare problem statement (1)

- Let's start with **<u>an easy example</u>**!
- The business use case description of a microwave oven is given
- Title of the BUC: "Microwave oven: how to heat food?"
- See next slide

Step 01 - Prepare problem statement (2)

“Microwave oven: how to heat food?”

ID	BUC_08, version v3.0	
Title	The heating of food	
Actor(s)	Cook	
Precondition(s)	The microwave oven is inactive, but it is plugged in The door is closed The food to heat is in close reach	
Normal flow	**Step**	**Description**
	01	The user opens the door
	02	The light goes on
	03	The user puts the food into the oven
	04	The user closes the door
	05	The light turns out
	06	The user presses the button once to set the working time to 60 seconds

Step 01 - Prepare problem statement (2)

“Microwave oven: how to heat food?”

Normal flow	Step	Description
(continued)	07	The light goes on
	08	Microwave tube starts
	09	The working time becomes visible on the display
	10	The working time decreases, and it is visible in the window. When the time is up:
	11	The light turns out
	12	A beep sounds
	13	The user opens the door
	14	The light goes on
	15	The user takes out the food
	16	The user closes the door
	17	The light turns out
Post condition	The food is heated	

Step 02 – Identify concepts, classes & objects (1)

"Microwave oven: how to heat food?"

Underline

ID	BUC_08, version v3.0	
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Step 02 – Identify concepts, classes & objects (2)

"Microwave oven: how to heat food?"

Normal flow	Step	Description
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Step 03 - Develop a common vocabulary (1)

- Develop a common vocabulary, dictionary, glossary
  - Count the occurrences

Nouns (1)	#	Nouns (2)	#	Nouns (3)	#
Door	4	Button	1	Display	1
Light	6	Working time	3	Window	1
Food	2	Seconds	1	Time	1
Oven	1	Microwave tube	1	Beep	1

  - Make an alphabetic list

Nouns (1)	#	Nouns (2)	#	Nouns (3)	#
Beep	1	Food	2	Seconds	1
Button	1	Light	6	Time	1
Display	1	Microwave tube	1	Window	1
Door	4	Oven	1	Working time	3

Step 03 - Develop a common vocabulary (2)

- Which domain concepts play an essential role?
- Consider whether concepts belong together
- Consider different words meaning same thing (synonyms)
- Consider same words possibly meaning something else (homonyms)
- Then
  - Annotate alphabetical list by indicating “Tangible”, “Concept”, “Event”, “External organization”, “Role”, “Other system”, “Interaction”, “Attribute”, “Structure”, “Device”, “Organizational unit”
  - Complete dictionary by adding definitions and comments

Step 03 - Develop a common vocabulary (3)

Nouns	#	Definitions and comments
Button	1	Tangible thing. Button is a critical concept.
Display	1	Tangible thing. Display is a critical concept.
Door	4	Tangible thing. Door is a critical concept.
Food	2	Interaction. Food is not part of the system, the oven …
Light	6	Tangible thing. Light is a critical concept.
Microwave tube	1	Tangible thing. Microwave tube is a critical concept.
Oven	1	Tangible thing. Oven tube is a critical concept. It is the “case” that keeps the whole together.
Seconds	1	Attribute. Unit of measure for working time.
Time	1	Concept. Time means the same as working time.
Window	1	Tangible thing. Window means the same a s display.
Working time	3	Concept. Working time is a critical concept. We need a clock, a timepiece

Step 03 - Develop a common vocabulary (4)

Nouns	#	Definitions and comments
Beep	1	Concept. Beep is a critical concept.
Button	1	Tangible thing. Button is a critical concept.
Display	1	Tangible thing. Display is a critical concept.
Door	4	Tangible thing. Door is a critical concept.
Food	2	Interaction. Food is not part of the system, the oven …
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Microwave tube	1	Tangible thing. Microwave tube is a critical concept.
Oven	1	Tangible thing. Oven tube is a critical concept. It is the "case" that keeps the whole together.
Seconds	1	Attribute. Unit of measure for working time.
Time	1	Concept. Time means the same as working time.
Window	1	Tangible thing. Window means the same a s display.
Working time (= clock)	3	Concept. Working time is a critical concept. We need a clock, a timepiece

Step 03 - Develop a common vocabulary (5)

- Create a glossary of terms, in collaboration with domain experts, developers, etc.
- In our example
  - We have 8 domain concepts
    (see previous slide)

- Each concept gets a descriptive statement
- Useful for developers who know little of the sector for which they must make the application

Step 03 - Develop a common vocabulary (6)

- Domain classes

- The domain concepts thus represent several objects …
- A class however **<u>is NOT the same as an</u>** object
  - A class is a *description* for similar objects
  - An object is a thing that is made according to the description
  - One class can have many objects

Example Pizza Perfect!

Step 03 - Develop a common vocabulary (7)

- So let us name the 8 domain classes of our example

Domain Class	Nouns	#	Definitions and comments
Alarm	Beep	1	Concept. Beep is a critical concept.
Button	Button	1	Tangible thing. Button is a critical concept.
Clock	Working time (= clock)	3	Concept. Working time is a critical concept. We need a clock, a timepiece
Display	Display	1	Tangible thing. Display is a critical concept.
Door	Door	4	Tangible thing. Door is a critical concept.
Light	Light	6	Tangible thing. Light is a critical concept.
Oven	Oven	1	Tangible thing. Oven tube is a critical concept. It is the “case” that keeps the whole together.
Tube	Microwave tube	1	Tangible thing. Microwave tube is a critical concept.

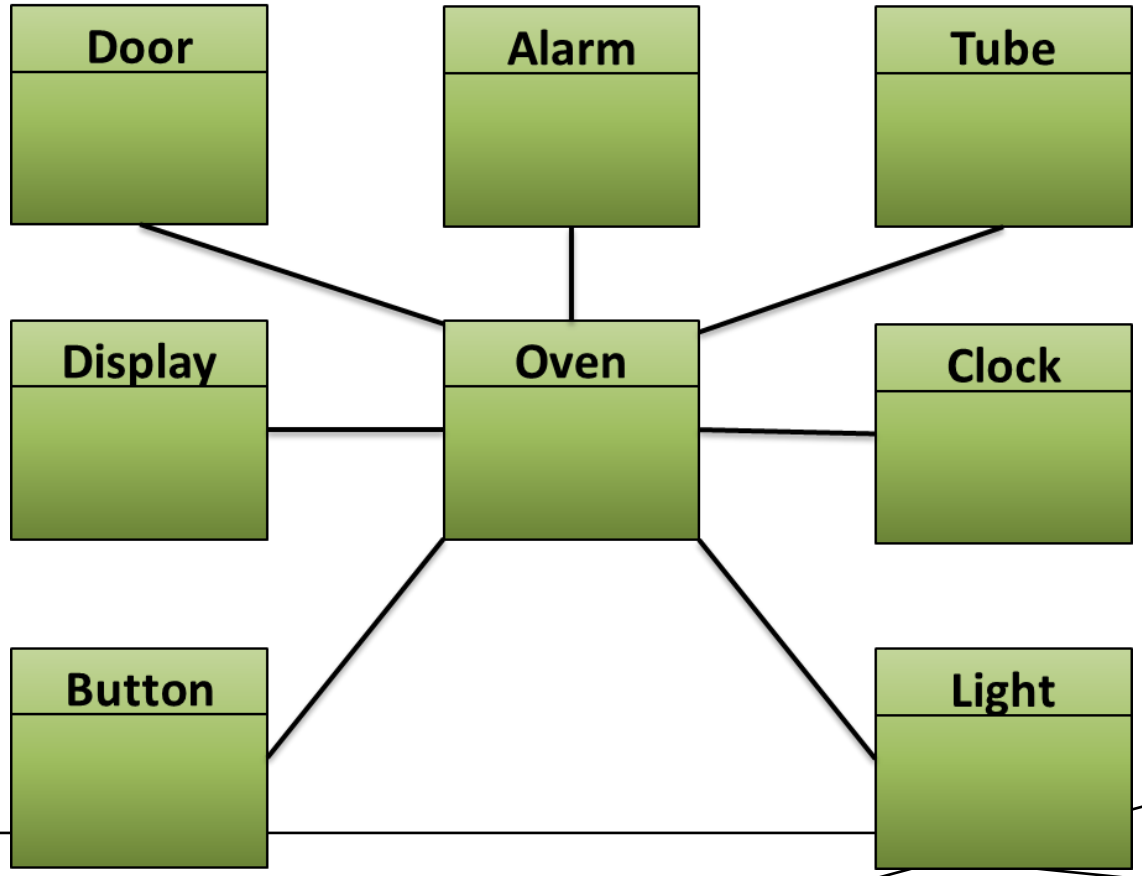
- **Further indebt analysis needed to clarify if there are more domain classes**

Step 03 – Glossary of terms (8)

Nouns	Explanation - description
Alarm	A short, typically high-pitched sound or tone, produced by the oven. It serves the purpose to notify when the timer is ended.
Button	The control element that allow you to operate the microwave oven and set parameters for cooking or heating food.
Clock	The clock on a microwave oven is a digital or analog timekeeping feature that displays the current time. It serves as a standard clock, similar to those found on alarm clocks or kitchen wall clocks, and has several functions and purposes like time display and timer.
Display	A digital or sometimes analog screen that provides information about the microwave's settings, cooking time, and other important details.
Door	A crucial safety feature that serves several purposes. It is a hinged or sliding panel on the front of the microwave oven that opens and closes to access the microwave's cooking chamber
Light	An interior light designed to illuminate the cooking chamber when the microwave is in operation.
Oven	The crucial kitchen appliance that uses electromagnetic radiation in the microwave frequency range to cook, heat, and reheat food quickly and efficiently. Microwave ovens are a common and convenient way to prepare meals, snacks, and beverages.
Tube	A high-powered vacuum tube that generates the microwave radiation used for cooking or heating food in a microwave oven.

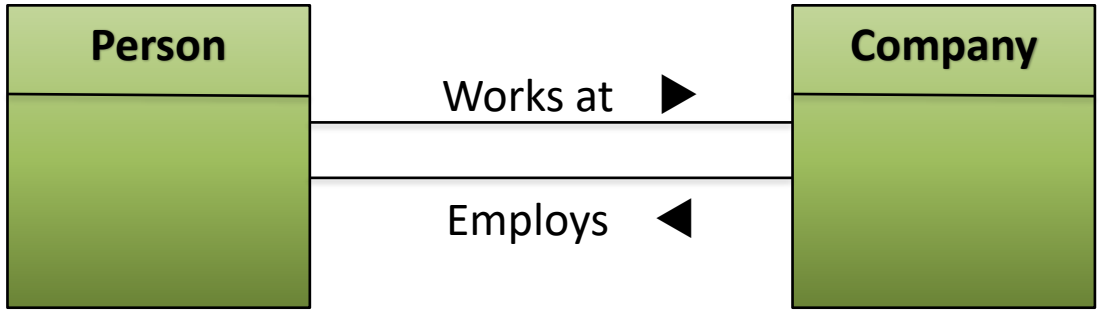
Step 03 - Develop a common vocabulary (8)

- Create a first **domain class diagram**
  - The oven is composed of the objects of the 7 other domain classes
  - "The oven is built of …."
  - The lines between the objects indicate that there is some kind of relationship between the domain classes



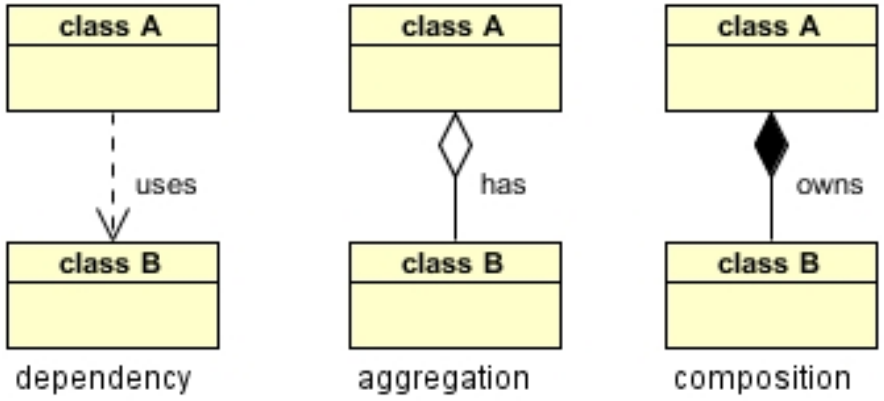
Step 04 - Associations between concepts (1)

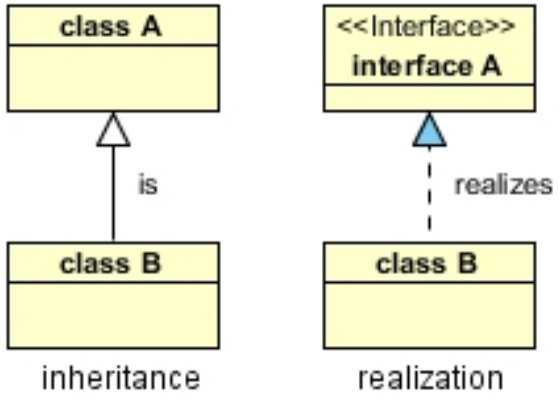
- Associations represent relationships between instances of classes
- For domain modeling associations represent **<u>conceptual relationships</u>**
  - Association = structural relation between 2 domain classes
    - Indicated by drawing a line between the 2 domain classes
    - Convention
      - Read names from left to right … but
      - Names can also be read from right to left (cf. arrow)
      - Example



Step 04 - Associations between concepts (2)

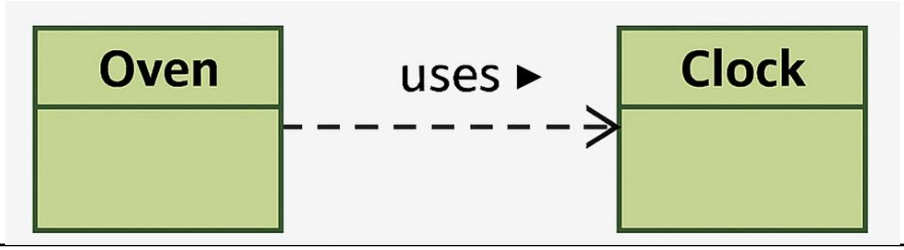
- The relationships are read as follows:
  - **Dependency:**
    - Class A uses class B
  - **Aggregation:**
    - Class A has a class B
  - **Composition:**
    - Class A owns a class B
  - **Inheritance:**
    - class B is a class A
      (or class A is extended by class B)
  - **Realization:**
    - Class B realizes class A
      (or class A is realized by class B)





Step 04 - Associations between concepts (3)

- Possible **conceptual relationships**
  - 🧩 **1. Dependency → "Uses temporarily"**
    - **Meaning**:
      - One class depends on another to do something, but only for a short time. If the other class changes, this one might break.
      - Think of this as a weak, short-term link.
    - ‼ **Example:**
      - The Oven depends on the Clock to show cooking time.
      - If the Clock design changes, the Oven's timer function may need updates.
      - 🗣 "The oven uses the clock, but the clock could exist without the oven."

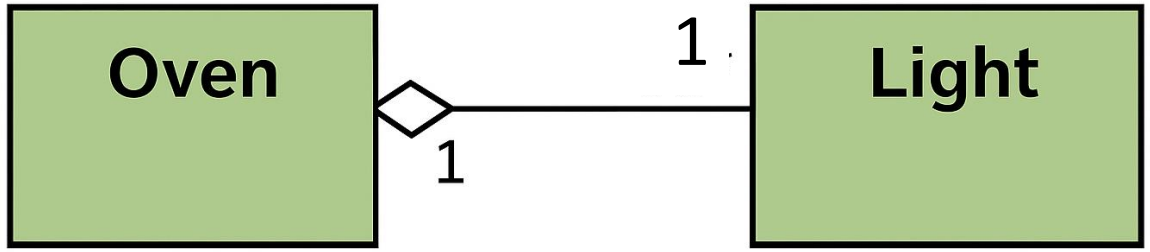


💡 **Tip:**
- **The oven depends on the clock.**
- **The clock does not depend on the oven.**
- **The oven needs the clock, but the clock doesn't need the oven."**

Step 04 - Associations between concepts (4)

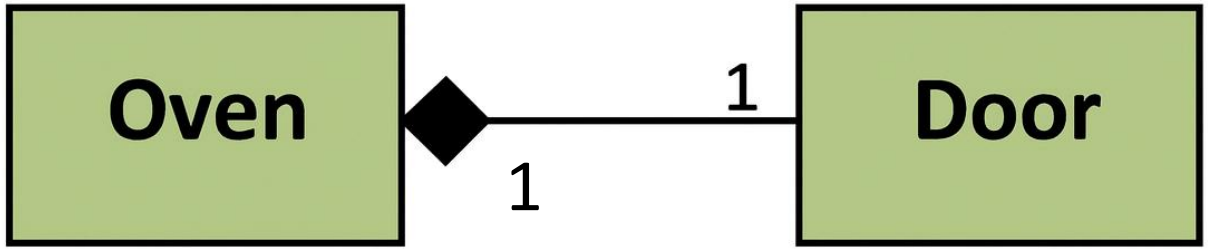
- Possible **<u>conceptual relationships</u>**
  - 🔗 **2. Aggregation → "Has a part, but can live without it“**
    - **Meaning:**
      - A whole–part relationship where parts can exist independently.
      - If the whole is destroyed, parts can still exist.
      - Represented by an empty diamond (◇).

    ‼ **Example:**
      - The Oven has a Light, but you could reuse the same Light in another oven model.
      - 🗣 “The oven has a light — but the light could belong to another oven too. In short when th



Step 04 - Associations between concepts (5)

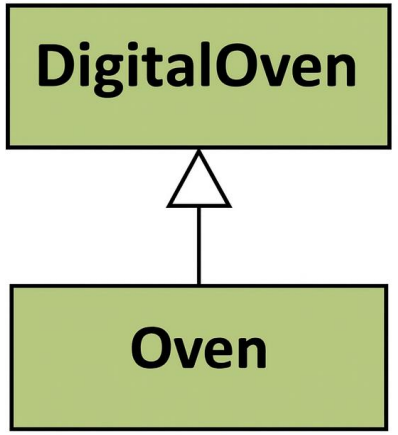
- Possible **<u>conceptual relationships</u>**
  - ⚙️ **3. Composition → "Part of me — can't live without me"**
    - **Meaning:**
      - A strong whole–part relationship.
      - If the whole is destroyed, the parts are destroyed too.
      - Represented by a filled diamond (◆).
    - ‼️ **Example:**
      - The Oven is made up of a Door, Tube, Display
      - If the oven is thrown away, those parts are gone too.
      - 🗣️ "The door only exists as long as the oven exists."



Step 04 - Associations between concepts (6)

- Possible **<u>conceptual relationships</u>**
  - 🧬 **4. Inheritance → "Is a type of“**
    - **Meaning:**
      - One class is a specialized version of another class.
      - The child class inherits the features of the parent.
      - Represented by a hollow triangle (△) arrow pointing to the parent.

    ‼ Example:
      - DigitalOven is a type of Oven.
      - It inherits all oven functions but adds extra features like a digital display.
      - 🗣 “A digital oven is a kind of oven.”



Step 04 - Associations between concepts (7)

- Going back to our example of the microwave oven
- Create based on the former slides the final solution for the microwave oven exercise
- Add the read direction and appropriate text

Step 05 - Assign attributes to the concepts (1)

- A model class consists out of attributes

- Attributes
  - Information or properties available for each object
  - These can be added to the domain model when it gives more clarity

- NOT included are the methods
  - Operations, actions, executed by the objects

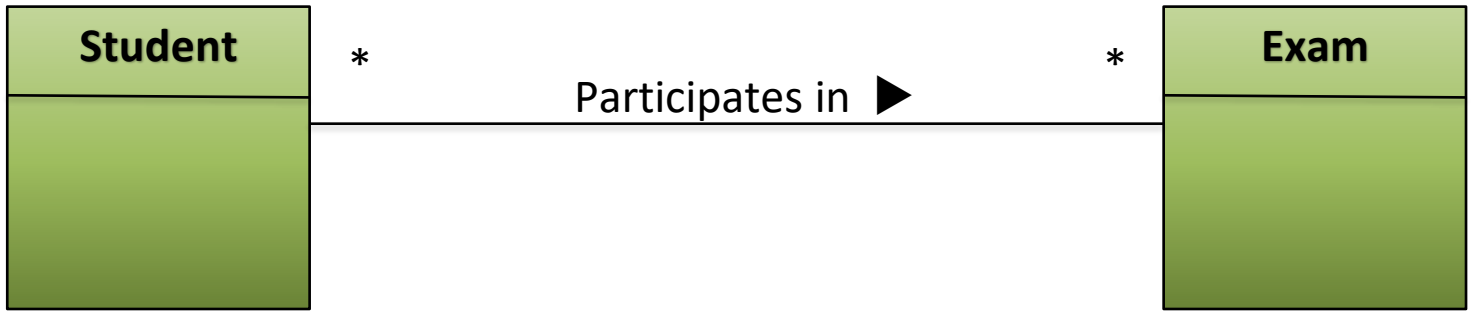
Remark: not covered in the example of the microwave oven

Step 06 - Check for multiplicities (1)

- For a better understanding of the association (s) between the domain classes
- Multiplicity is the number of instances of a particular class that is involved in an association in relation to exactly one instance of another class

Step 06 - Check for multiplicities (2)

- For example,



- From left to right
  - Each student will participate in an indeterminate (* right) number of exams
- From right to left
  - Each exam is attended by an indefinite (* left) number of students

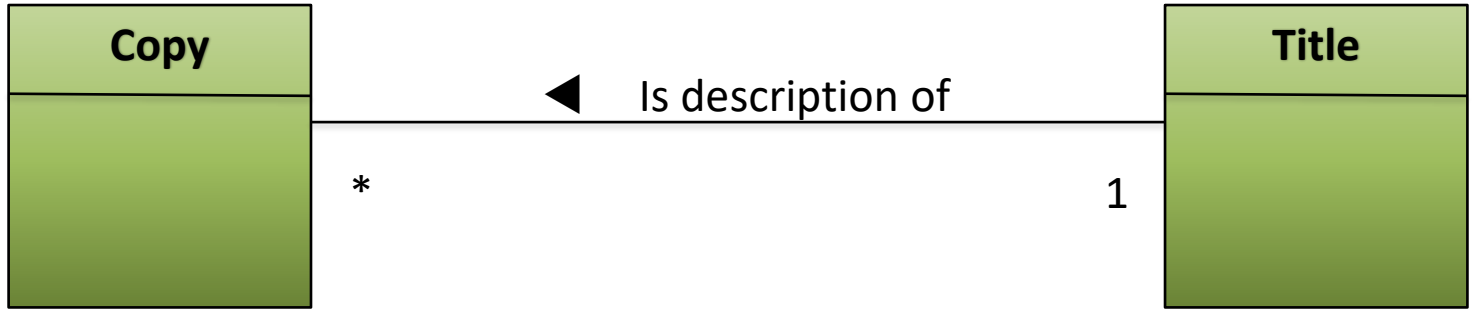
Step 06 - Check for multiplicities (3)

- Multiplicities in UML

Multiplicity	
1	Exactly 1
2	Exactly 2
1 .. 5	From 1 to 5 (included)
6, 7	6 or 7
1 .. *	Undefined number but at least 1
0 .. *	Undefined number, eventually 0 Is the same as *
*	Undefined number, eventually 0 Is the same as *
0 .. 1	0 or 1 Is the same as 0, 1

Step 06 - Check for multiplicities (4)

- Example - Library



- To every copy there only belongs one title
- For every title there exists an indefinite number (possibly 0) copies

Step 06 - Check for multiplicities (5)

- Example - Chess

- Every chessboard consists out of 64 fields
- Every field belongs to exactly one chessboard

Exercises – University system

- In the beginning of each semester, students can select and register different courses which are thought by different teachers. Each student is uniquely identified by his or her student ID. Apart from student ID, student name. dob (date of birth), mobile number, e-mail ID are also stored in the university records. Like students, teachers also have a unique teacher ID and other teacher attributes like teacher's name, specialization, et cetera. These are also stored in the system. Each semester, a student must register at least three subjects and five at most. The student may participate in different activities. The courses have a course name and a course ID. Each teacher must teach at least one subject in each semester
- Create the full domain model with glossary of terms

Exercises

- MS Word doc
  - ABO De Lijn
  - Tank station
  - Placing an order
  - Online Stock

Questions & answers